SUPERSEDING
Int. Fed. Spec. J-C-00129b (GSA-FSS)
August 24, 1961 and
Fed. Spec. J-C-129a
September 15, 1955

FEDERAL SPECIFICATION

CABLE AND WIRE; THERMOPLASTIC—INSULATED, GENERAL PURPOSE (0—TO 600—VOLT SERVICE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single conductor cable and wire employing soft-annealed-copper or aluminum conductors, solid or stranded, with a flame-retardant thermoplastic insulation, having no fibrous covering, for use at 600 volts or less.

1.2 Classification.

1.2.1 *Types, classes, and kinds.* Cable and wire furnished under this specification shall be of the following types, classes, and kinds, as specified (see 6.1).

Types:

TW—Moisture resistant insulation, 60° C. operation—wet or dry.
THW—Heat and moisture resistant insulation, 75° C. operation—wet or dry.

Classes:

Class 1—Solid conductor (AWG 14-AWG 8 only).
Class 2—Stranded conductor of standard flexibility, conforming to class B of ASA C7.8 or C7.21.
Class 3—Stranded conductor of greater flexibility, conforming to class C of ASA C7.8 or C7.21.
Class 4—Stranded conductor of extreme flexibility, conforming to class G of ASA C7.13.

Kinds:

Kind A—Aluminum conductor.
Kind C—Copper conductor.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

J-C-98—Cable and Wire, Insulated; Methods of Sampling and Testing.
UU.P-271—Paper, Wrapping, Waterproof, Kraft.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Box, Fiberboard.
PPP-B-636—Boxes, Wood, Nailed and Lock Corner.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies)

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547—Liner, Case, Waterproof.
MIL-C-12000—Cable, Cord and Wire, Electric Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109—Inspection Terms and Definitions.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

Underwriters' Laboratories, Inc. Standards:

UL83—Standard for Thermoplastic—Insulated Wires.

(Copies may be obtained from the Underwriters' Laboratories, Ins., 161 Sixth Avenue, New York 13, New York, 207 East Ohio Street, Chicago, Ill., or 1655 Scott Blvd., Santa Clara, California.)

American Standards Association, Inc. Standards:

C7.8—Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft, Specifications for.

C7.13—Rope-Lay-Stranded Copper Conductors Having Concentric-Stranded Members, for Electrical Conductors, Specifications for.

C7.21—Concentric-Lay-Stranded Aluminum Conductors, Hard-Drawn, Three-Quarter Hard-Drawn, and One-Half Hard-Drawn, Specification for.

(Copies may be obtained from the American Standards Association, Inc., 10 East 40th Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Cable and wire covered by this specification shall conform to the requirements of Underwriters' Laboratories, Inc. Standard UL83 For Thermoplastic-Insulated Wires and the pertinent American Standards Association, Inc. standards set forth in 2.2.

3.1.1 *Conductor size.* Conductors in cable and wire furnished under this specification shall be of the following sizes, as specified in the invitation for bids.

Cable and Wire Type	*AWG or circular MIL size of conductor*
TW	*14 to 2,000,000
THW	*14 to 500,000

*For soft-annealed-copper conductors only.

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Aluminum conductor sizes are permissible from No. 12 AWG to 2,000,000 circular MIL for type TW insulation and from No. 12 AWG to 500,000 circular MIL for type THW insulation.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the cables and wires he proposes to supply under this specification conform to the requirements of Underwriters' Laboratories, Inc. Standard UL 83 for Thermoplastic-Insulated Wires. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the cables and wires conform to these requirements.

3.2.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the contracting agency that his cables and wires conform to the published standards including methods of tests of the Underwriters' Laboratories, Inc. Standard UL 83 for Thermoplastic-Insulated Wires.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier's responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Unless otherwise specified, such inspections by the supplier may be performed on a statistical or production lot basis to assure that the quality of manufacture is equal to the standards set by Underwriters' Laboratories, Inc. (See 6.2).

4.2 Sampling, nonmilitary.

4.2.1 *Lot.* Unless otherwise specified in the invitation for bids, the lot shall be formed as described in paragraph 3 of section 6 of J-C-98.

4.2.2 *Sampling for inspection and tests.* Unless otherwise specified in the invitation for bids, the sample shall be taken as described in paragraph 6 of section 6 of J-C-98.

4.3 Visual inspection, nonmilitary.

4.3.1 Each inspection unit in the sample, 4.2.2, as well as the package from which the inspection unit is taken, shall be examined visually before any tests are made. Visual inspection shall cover construction, tinning of conductors, splices, marking, type of conductor, joints, and repairs, and may be limited to that part of the package which is visible without unrolling.

4.4 General inspection, Military.

4.4.1 *Definitions.* Definitions of inspection terms used herein shall conform to ML-STD-109.

4.4.2 *Acceptance inspection.* Acceptance

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inspection shall consist of groups A, B, and C inspection in 4.4.2.1 through 4.4.2.3.1. Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection or both.

4.4.2.1 *Group A inspection.* Group A inspection including sampling shall be in accordance with table I and MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

TABLE I. *Group A inspection*

Inspection	AQL	
	Major	Minor
Visual and Mechanical:	*Percent*	*Percent*
General construction Tinning Splices Marking Type of conductor (diameter) Joints Repairs	1.0	4.0
Electrical:		
Insulation resistance	1.0**	
Dielectric strength		*

*All electrical defects are considered major.
**1.0 percent unless otherwise specified.

4.4.2.2 *Group B inspection.* This inspection including sampling shall conform to table II and to MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1.

TABLE II. *Group B Inspection*

Inspection	Sampling plan
Average thickness of insulation Minimum thickness of insulation (less than nominal 3/64 inch) Minimum thickness of insulation (nominal 3/64 inch and over) Lay of pitch, lay of twist	See 4.4.2.2.1
Accelerated aging, air oven test	6 specimens per 100,000 feet
Flame retardant properties, insulation	6 specimens per 100,000 feet or fraction thereof
Moisture absorption, insulation	6 specimens per 100,000 feet or fraction thereof

4.4.2.2.1 *Group B sampling plans.* The group B sampling plans listed in table II shall be as follows:

AQL	Inspection level for normal inspection	Inspection level for reduced inspection
Percent		
6.5	L-7	L-5

4.4.2.2.2 *Order of inspection within Group B.* Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.4.2.2.3 *Procedure in the case of failure.* If an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

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4.4.2.3 *Group C inspection.* This inspection shall be as listed in table III.

TABLE III. *Group C inspection*

Inspection	Sampling plan
Flexibility Deformation Tensile strength, insulation and neoprene sheath Elongation, insulation Specific inductive capacity	6 specimens per 500,000 feet

4.4.2.3.1 *Specimens.* Six specimens shall be prepared and tested from each lot of 500,000 feet or fraction thereof of cable and wire. If each of the 6 specimens meet the specified requirements, the lot shall be accepted.

4.4.2.3.1.1 If 2 or more of the 6 specimens fail to meet the specified requirements, the lot shall be rejected.

4.4.2.3.1.2 If only 1 of the 6 specimens fails to meet the specified requirements, a second sample of 6 specimens shall be tested. Lot acceptance shall be permitted only if in the combined sample of 12 specimens not more than 1 specimen fails to meet the requirements. Failure of 2 or more specimens in the combined first and second samples shall result in lot rejection.

4.4.2.3.2 *Noncompliance.* If a lot fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and on all units of product which were manufactured with the same conditions, materials, processing, etc. If the Government inspector does not consider that the corrective action will enable the product to meet the specified requirements or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3 *Disposition of nonconforming product.* Disposition of lots and sample units rejected during group A or B inspection shall be in accordance with MIL-STD-105. The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means to correct the fault:

(*a*) Sample units found defective during groups A, B, or C inspection.

(*b*) Inspection lots rejected during group A or B inspection.

(*c*) Product has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Tests.

4.5.1 *Nonmilitary.* The wire and cable shall be subjected to the tests set forth in Underwriters Laboratories, Inc. Standard UL83 to determine compliance with 3.1.

4.5.2 *Military.* The wire and cable shall be subject to the tests set forth in Underwriters Laboratories, Inc. Standard UL83, modified by the instructions in 4.4, to determine compliance with 3.1.

4.6 Compliance criteria, nonmilitary.

4.6.1 *Inspection.* If one or more of the packages, 4.2.2, fails to conform to the requirements for which inspection was made, 4.3, the lot from which the packages were taken shall be considered to have failed to meet the requirements of this specification.

4.6.2 *Tests.* Compliance of the lot with respect to tests shall be based on the principles described in section 6 of J-C-98.

4.6.2.1 Any reel, coil, or roll of wire or cable which shows the presence of a defect or fails to meet the requirement when sub-

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jected to the electrical tests shall be considered to have failed to meet the requirements of this specification. If after the defects are repaired and the roll, reel, or coil is again subjected to the dielectric strength and insulation resistance tests, the presence of a defect is not shown, the roll, reel, or coil shall be considered to have met the requirements of this specification for dielectric strength and insulation resistance.

4.7 Compliance criteria, Military.

4.7.1 The compliance criteria shall be as set forth in 4.4.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the levels of preservation, packaging and packing protection used are defined in Fed. Std. No. 102.)

5.1 Packaging.

5.1.1 *Level A and B.*

5.1.1.1 *Coils (sizes AWG 14 to AWG 10).* Cable and wire shall be uniformly coiled and packaged 500 feet in a fiberboard box conforming to PPP-B-636, type I or type II, class 2, provided with the special knock-out feature normally used in commercial practice for dispensing purposes.

5.1.1.2 *Coils (sizes AWG 8 to AWG 2).* Cable and wire shall be uniformly coiled and packaged 500 feet per coil. The coil shall be spirally wrapped with a waterproofed kraft paper conforming to UU-P-271, class E-2 and sealed.

5.1.1.3 *Reels.* Cable and wire shall be packaged on non-returnable wood or metal reels suitably designed with strong flanges of sufficient diameter to protect the cable or wire from damage during shipment and storage. After the cable or wire has been wound on the reel, it shall be covered with a full wrap of waterproofed kraft barrier material conforming to UU-P-271, class E-2. The wrap shall extend between the flanges and shall be secured in place with a minimum of two flat steel straps.

5.1.2 *Level C.* Cable and wire shall be packaged in coils or on reels of the size and kind commonly used which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.1.3 *Military agencies — Levels A and B.* The cable and wire shall be packaged in accordance with level A or B, as applicable, in accordance with MIL-C-12000.

5. 2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Coils (sizes AWG 14 to AWG 10).* Cable and wire packaged as specified in 5.1.1.1 shall be packed in nailed wood or wood cleated plywood boxes conforming to PPP-B-621 and PPP-B-601, respectively. Containers shall be fitted with waterproof case liners conforming to MIL-L-10547, class 2 and sealed in accordance with the appendix thereto.

5.2.1.2 *Coils (sizes AWG 8 to AWG 2)* and reels. Cable and wire packaged as specified in 5.1.1.2 and 5.1.1.3 shall require no further overpacking.

5.2.2 *Level B.*

5.2.2.1 *Coils and reels.* Cable and wire packaged as specified in 5.1.1.1, 5.1.1.2 and 5.1.1.3, shall require no further overpacking.

5.2.3 *Level C.* The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.2.4 *Military agencies — Levels A and B.* Cable and wire shall be packed in accordance with level A or B, as applicable, in accordance with MIL-C-12000.

5.3. Marking.

5.3.1 *Civil agencies.* In addition to any special marking specified in the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking specified in the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.
(b) Type, class, and kind of cable (1.2.1).
(c) Size of cable (3.1.1).
(d) Administrative provisions for inspection records (4.1).
(e) Lot, if other than specified (4.2.1).
(f) Sampling, if other than specified (4.2.2.).
(g) Levels of packaging and packing required (5.1 and 5.2).
(h) Marking required (5.3).

6.2 If laboratory tests by the Government are required, purchasing officers should order wire or cable for test purposes in addition to the number of feet required for installation.

6.3 Underwriters' Laboratories, Inc. Standard No. 83 for Thermoplastic-Insulated Wires permits a substitution of the spark test in place of part of dielectric strength and insulation resistance tests for 600 volt wire and cable. Since the spark test is usually conducted as a part of the manufacturing process, the dielectric strength and insulation resistance (when required) should be specified in the invitation for bids and arrangements should be made with the manufacturer for witnessing the tests in his plant.

6.4 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Rail — Copper
Cable, electric, copper, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor
Cable, electric, copper, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 34, National Motor Freight Classification.

Rail — Aluminum
Cable, electric, aluminum
Carload minimum weight 30,000 pounds.

Motor
Cable, electric, aluminum
Truckload minimum weight 30,000 pounds, subject to Rule 34, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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